Your six-hour study route

Exam tomorrow, 15 October 2026 / 360 minutes including breaks

Main conclusion: prepare formulas, explanations, and multi-step calculations. Quiz 1 directly tests GAN gradient reasoning and Inception Score arithmetic. It supports this question style; it cannot establish the midterm topic weights or exact paper format.

Minutes	Work	Result you must produce without notes
0-50	GAN losses and training	Derive both generator logit gradients; explain freezing and mode collapse.
50-100	Evaluation metrics	Redo Quiz 1 IS; calculate diagonal FID and kNN precision/recall.
100-110	Break	Stop studying for ten minutes.
110-150	Introduction + AE/VAE	Bayes posterior, AE vs VAE, Gaussian KL, reparameterized sample.
150-210	Diffusion Parts 1-2	Forward sample, alpha products, training target, one reverse step.
210-220	Break	Stop studying for ten minutes.
220-280	Transformers	Q/K/V attention, causal mask, positional encoding and cross-entropy.
280-340	Closed-book mock	Attempt pages 8-9 in 60 minutes; no answer checking mid-attempt.
340-360	Mark and repair	Use pages 10-11; redo your two weakest calculations.

Use your existing material efficiently

The handwritten-notes folder already has 37 revision pages covering all seven decks. Use its README reading order. The older support.md and numerical-practice.md concentrate on Introduction, AE/VAE and GANs; they do not replace diffusion or Transformer preparation.

For each block: read a compact explanation, close it, solve a problem, then explain the answer aloud. Spend little time copying notes or reading repeated slide builds. Study priorities here reflect the quiz and the mathematical emphasis of the slides, not a claimed exam prediction.

What Quiz 1 tells us

Observed evidence / 20 minutes / 8 marks / two questions worth 4 marks each

Q1: an equation plus a causal explanation

The quiz asks for saturating and non-saturating GAN losses, why non-saturating is preferred, and what happens when D(G(z)) is near zero. A strong answer gives both formulas and then explains the gradient through the sigmoid.

```
Let d = D(G(z)) = sigmoid(a).
L_sat = E_z[ln(1-d)] ; L_NS = -E_z[ln d]
dL_sat/da = -d ; dL_NS/da = d-1
```

As d approaches zero, the saturating loss approaches 0 but its logit gradient approaches 0; the non-saturating loss grows large while its logit gradient approaches -1. Minimizing either pushes the fake logit upward. Non-saturating loss therefore avoids this particular early-training saturation.

Precision point: these derivatives are with respect to the discriminator logit a, not directly to d or all generator parameters. The full generator gradient also includes da/d(theta_G). A strong logit gradient alone does not guarantee successful training.

Q2: reproduce the full calculation

Image	Class probabilities	KL to marginal
1	(0.80, 0.15, 0.05)	0.52792841
2	(0.02, 0.95, 0.03)	0.34790573
3	(0.10, 0.70, 0.20)	0.14827437

```
p(y) = (0.30666667, 0.60000000, 0.09333333)
KL_i = sum_c p(c|x_i) ln[p(c|x_i)/p(c)]
mean KL = 0.34136950 ; IS = exp(mean KL) = 1.40687299
```

Show the column means, one weighted log-ratio per class, each row sum, the mean across images, and one final exponentiation. Do not divide each KL by the number of classes. Keep precision until the final answer, about 1.41.

Reasonable midterm question styles

Expect to prepare for: write-and-explain objectives; derive a short formula; calculate from supplied vectors or matrices; interpret a score or failure case; compare training and inference; label a model architecture. These are preparation targets inferred from the quiz and lectures, not confirmed midterm questions.

Only the quiz has an observed 50/50 conceptual-numerical split. There is no evidence that the midterm will preserve that split, its marks, or its duration.

Foundations, AE and VAE

Sources: Introduction pp. 3-4, 15-41; AE/VAE pp. 5-19, 20-64

What you need to explain

Generative modeling fits a data distribution to sample plausible new observations. Distinguish p_data from p_model, sample count N from feature dimension M, and density estimation from generation. Discriminative classification models p(y|x); a class-based generative approach can model p(x|y)p(y) and recover the posterior with Bayes. An unconditional generator need not model labels.

```
P(y|x) = P(x|y)P(y) / sum_k P(x|k)P(k)
```

Lecture spam example: scores 0.60 x 0.40 = 0.24 and 0.05 x 0.60 = 0.03; normalized spam posterior = 0.24/0.27 = 0.888889. A raw likelihood-times-prior score is not yet a posterior.

Autoencoder: reconstruction first

Draw x -> encoder -> z -> decoder -> x_hat. An AE learns a deterministic representation by reconstruction. Plain AE training does not impose a prior that makes arbitrary latent sampling reliable. Know compression, retrieval via latent distance, denoising, colorization, and anomaly detection.

Denoising input/target: corrupted image / clean image. Colorization: grayscale / color. Anomaly detection uses a reconstruction-error threshold; it is a heuristic and can fail. Distinguish sum squared error from mean squared error before substituting values.

VAE: a distribution for each input

```
q_phi(z|x) = N(mu, diag(sigma^2)); p(z) = N(0,I)
logvar = ln(sigma^2); sigma = exp(0.5 logvar)
z = mu + sigma * epsilon ; epsilon ~ N(0,I)
KL(q||p) = 0.5 sum_j[mu_j^2 + sigma_j^2 - 1 - ln(sigma_j^2)]
```

Minimize negative ELBO: expected negative log-likelihood of reconstruction plus KL(q_phi(z|x)||p(z)). A beta-VAE changes the KL coefficient. Explain reconstruction fidelity versus pressure toward the prior; regularization encourages smoother sampling and interpolation, without guaranteeing every decoded point is meaningful.

Derivation and numerical targets

Derive Gaussian KL from E_q[ln q - ln p], using E[(z-mu)^2] = sigma^2 and E[z^2] = sigma^2 + mu^2. Reparameterization puts randomness in epsilon and leaves a differentiable function of mu and sigma; holding epsilon fixed, dz/dmu = 1 and dz/dsigma = epsilon.

Practice: mu=(1,0), variances=(4,0.25), epsilon=(0.5,-2). Standard deviations=(2,0.5); z=(2,-1); KL=1.625. If reconstruction NLL is 0.2, negative ELBO is 1.825. KL is zero for mu=0 and variance=1, and cannot be negative.

Likely-style prompts: Why is a VAE more suitable for latent sampling than a plain AE? Derive Gaussian KL. Compute a sample from logvar. Explain why direct random sampling is awkward for ordinary backpropagation.

GANs and conditional generation

Sources: GANs Part 1 pp. 28-55, 59-69; GANs Part 2 pp. 2-22

Objectives and an alternating update

```
V(D,G) = E_real[ln D(x)] + E_z[ln(1-D(G(z)))]
D maximizes V, or minimizes L_D = -V.
G minimizes L_sat = E_z[ln(1-D(G(z)))]
Practical G loss: L_NS = -E_z[ln D(G(z))]
```

D step: use real labels 1 and fake labels 0; detach the generator output and update D. G step: generate fakes, use real targets for their D scores, freeze D parameters, and backpropagate through D into G. Freezing D weights must not disable the gradient path through its input.

Numerical check with one real/fake pair: D(x)=0.8 and D(G(z))=0.1. Using the sum of the two terms, L_D=-ln(0.8)-ln(0.9)=0.328504; L_sat=ln(0.9)=-0.105361; L_NS=-ln(0.1)=2.302585. If a question averages the two D examples together, its D loss is half this sum.

Training failures: give mechanism, symptom, remedy

Vanishing gradient: D confidently rejects fakes, weakening the saturating G gradient; use non-saturating G loss. Mode collapse: many latent inputs generate a narrow set of outputs; batch diversity features/minibatch discrimination help D detect repetition. This is different from vanishing gradients and is not automatically fixed by changing G loss.

Oscillation: coupled G/D updates can keep changing each other's targets. Discuss learning-rate adjustment, update balance and two learning rates. Loss alone is an unreliable image-quality measure; inspect samples, diversity and evaluation metrics. No single G/D update ratio is universally best.

Architecture and latent behavior

Know the DCGAN picture: noise is projected and reshaped, then upsampled through convolutional stages; D reduces an image to a real/fake score. Read spatial dimensions, channels and the final output shape. Explain latent interpolation and vector arithmetic as empirical properties, not guaranteed semantic laws.

Conditional GAN and StackGAN

A cGAN supplies condition y to both G(z,y) and D(x,y); D checks the image-condition pair. Conditions can be class one-hot vectors, text embeddings or images. If noise has 100 coordinates and a label is a 10-way one-hot vector, concatenated G input has 110 coordinates. For multiple categorical fields, concatenated width is the sum of category counts, while possible combinations multiply.

StackGAN Stage I draws a 64x64 sketch from text conditioning and noise. Stage II refines the sketch plus text into 256x256 detail. Conditioning augmentation samples around an encoded text condition and applies KL regularization. Stage II has no separate fresh latent z, but conditioning augmentation can still introduce randomness. Know GAN vs VAE tradeoffs: sharpness, stability, latent sampling, mode coverage and inference cost.

Likely-style prompts: Trace which weights and gradients change in each update. Diagnose high-quality repeated samples. Why condition D as well as G? Explain the two-stage motivation.

Evaluation: calculate and interpret

Source: GANs Part 2 pp. 24-96 / Direct Quiz 1 overlap: Inception Score

Inception Score: confidence and class diversity

```
m_c = (1/N) sum_i P_ic
IS = exp[(1/N) sum_i sum_c P_ic ln(P_ic/m_c)]
ln IS = H(Y) - H(Y|X)
```

High IS rewards confident per-image predictions and diverse predicted classes across the set. It does not compare against real reference images or establish novelty. If every prediction vector is identical, IS=1, even with confident predictions. Repeating one prototype in each of several classes can still score well. Higher is preferred under a fixed classifier and protocol.

For the entropy derivation, expand ln(P_ic/m_c) into ln P_ic - ln m_c. Averaging the second term uses the column means m_c; the result is marginal entropy minus mean conditional entropy. Remember zero-probability terms contribute zero by a limit.

FID: difference in feature means and covariances

```
FID = ||mu_r-mu_g||^2 + Tr[S_r + S_g - 2(S_r S_g)^(1/2)]
For diagonal S: covariance term = sum_j(sqrt(v_rj)-sqrt(v_gj))^2
```

Lecture example: mu_r=(1,2), mu_g=(2,3), S_r=diag(2,2), S_g=diag(3,3). Mean term=2; covariance term=10-4 sqrt(6)=0.202041; FID=2.202041. Trace means sum the diagonal entries. These inputs are variances, not standard deviations.

Lower FID is preferred with the same feature extractor, preprocessing and sample protocol. FID=0 establishes matching fitted Gaussian moments, not identical arbitrary image distributions or no memorization. The square root is a matrix square root; elementwise roots work here because the matrices are diagonal.

Improved generative precision / recall

For every point, compute its k-th nearest other neighbor in its own set; this distance is its ball radius. Exclude self. A query belongs to the estimated manifold if any reference ball covers it, including the boundary.

```
Precision = generated points inside real balls / number generated
Recall = real points inside generated balls / number real
```

High precision + low recall suggests realistic but narrow generation. Low precision + high recall suggests broad coverage with poor fidelity. Neither is ordinary label-based classification precision/recall. Test the union of balls: checking only the nearest center is not generally sufficient.

Lecture checkpoints, k=1: real radii=(1,1,1.118034,1.5,1.5); generated radii=(0.538516,0.5,0.538516,0.5,1.984943). Four of five generated and three of five real points are covered: precision=0.8, recall=0.6.

Source trap: the slide claim that confident outputs of only one class score well on IS is inconsistent with the formula; IS=1 for identical predictions. FID also does not automatically detect copying. Use the definitions when explaining these cases.

Diffusion: forward, train, reverse

Sources: Diffusion Part 1 pp. 14-34; Part 2 pp. 4-39 and Appendix A pp. 43-46

Forward noising and its derivation

```
alpha_t = 1-beta_t ; alpha_bar_t = product_(s=1..t) alpha_s
x_t = sqrt(alpha_t)x_(t-1) + sqrt(beta_t)epsilon_t
x_t = sqrt(alpha_bar_t)x_0 + sqrt(1-alpha_bar_t)epsilon
q(x_t|x_0) = N(sqrt(alpha_bar_t)x_0, (1-alpha_bar_t)I)
```

For independent unit-variance inputs, Var(a x + b epsilon)=a^2+b^2. Setting b^2=beta and a^2=1-beta explains the square roots. The conditional variance given a fixed clean image is 1-alpha_bar_t; do not confuse this with a unit-variance marginal assumption. Naive independent additive noise grows variance and does not attenuate the signal mean.

Derive the closed form by substituting one previous step and adding independent Gaussian variances: alpha_t(1-alpha_bar_(t-1)) + beta_t = 1-alpha_bar_t. The single epsilon is aggregate noise, not necessarily the last incremental epsilon from a sequential trajectory.

Lecture 2x2 example: beta=(0.19,0.36,0.51); alpha=(0.81,0.64,0.49); alpha_bar=(0.81,0.5184,0.254016). Thus x_3=0.504 x_0 + 0.86370365 epsilon. With x_0=[[0.8,-0.6],[0.4,1]] and epsilon=[[1,-0.5],[0.5,-1]], x_3=[[1.266904,-0.734252],[0.633452,-0.359704]]. Noisy values may leave [-1,1].

Training and schedules

```
L_simple = E_(x0,t,epsilon)[||epsilon-epsilon_theta(x_t,t)||^2]
```

Sample a clean image, random timestep and Gaussian noise; form x_t directly; pass (x_t,t) into the U-Net; regress to the noise added in the closed form. Training does not run a full reverse trajectory. The model predicts aggregate noise, not just the last increment. Time conditioning tells it the noise level.

Linear scheduling makes beta linear, not alpha_bar. A normalized cosine choice is alpha_bar_t=f(t)/f(0), f(t)=cos^2(((t/T+s)/(1+s)) pi/2); beta_t=1-alpha_bar_t/alpha_bar_(t-1), with a cap. Dropping timestep-dependent weights produces the simplified objective; it is not an algebraically identical ELBO.

Reverse sampling and the derivation target

```
x0_hat = [x_t-sqrt(1-alpha_bar_t)epsilon_theta]/sqrt(alpha_bar_t)
mu_theta = [x_t-beta_t epsilon_theta/sqrt(1-alpha_bar_t)]/sqrt(alpha_t)
x_(t-1) = mu_theta + sigma_t z ; z~N(0,I), or z=0 at t=1
```

Start inference from x_T~N(0,I), reuse the denoiser for T backward steps, and omit added noise in the final step. For Appendix A, substitute x0_hat into posterior mean A x_0+B x_t, where A=sqrt(alpha_bar_(t-1))beta_t/(1-alpha_bar_t) and B=sqrt(alpha_t)(1-alpha_bar_(t-1))/(1-alpha_bar_t). Collect terms to obtain the displayed reverse mean. If using posterior variance, sigma_t^2=beta_t(1-alpha_bar_(t-1))/(1-alpha_bar_t).

Transformers: the numerical route

Source: Transformer pp. 8-29, 30-58, 59-67 / Work in radians

Architecture and scaled dot-product attention

Know embedding lookup plus positional information; encoder self-attention, FFN, residual connection and LayerNorm; decoder causal self-attention, cross-attention, FFN; final vocabulary projection and softmax. Multiple heads learn different projections, concatenate their outputs, then apply an output projection. FFN acts separately at each token.

```
Using row vectors: Q=XW_Q ; K=XW_K ; V=XW_V
Attention(Q,K,V) = softmax_rows(QK^T/sqrt(d_k) + mask)V
```

Procedure: project Q/K/V, form query-key dot products, divide by sqrt(d_k), mask invalid scores, softmax across keys, then take the weighted sum of value vectors. Explain scaling as control of growing dot-product variance that can saturate softmax. For n tokens, scores have shape n x n; a head output has n x d_v.

Encoder self-attention: Q/K/V come from the encoder sequence. Causal decoder self-attention: current and earlier input positions are allowed; future positions receive -infinity before softmax. Cross-attention: Q comes from the decoder; K/V come from encoder outputs. Cross-attention does not use a triangular mask over source positions, though padding may be masked.

Positional encoding and training

```
PE(pos,2i) = sin(pos / 10000^(2i/d_model))
PE(pos,2i+1) = cos(pos / 10000^(2i/d_model))
CE_sum = -sum_t ln p_t(correct token) ; CE_mean=CE_sum/T
```

For d_model=10, pos=3, i=0..4, the vector is (0.141120,-0.989992,0.457755,0.889079,0.075285,0.997162,0.011943,0.999929,0.001893,0.999998). Both coordinates in a pair use the same denominator. At pos=0 the vector alternates 0,1.

Teacher forcing uses shifted ground-truth decoder inputs: [SOS,y1,y2,...] predicts [y1,y2,y3,...]. At inference, feed generated tokens back until EOS or a length limit. Greedy decoding chooses the best next token; beam search retains several partial sequences using cumulative log probability. Beam search is approximate, not a guarantee of a globally best sentence.

Exact lecture numerical checkpoints

Robot, pp. 21-22: Q_robot=(1.25,1.85); raw scores=(5.9775,3.905,1.836). With exact scaling and softmax, weights=(0.778546,0.179819,0.041635); output=(5.606850,2.187793). The displayed slide solution has an arithmetic error. Use a consistent row-vector convention with the provided matrices.

Cross-attention, p. 58: query=(5,3.41); weights=(0.002414,0.047216,0.923457,0.026912); output=(3.748287,1.465033). Sentence CE, pp. 66-67: correct-token probabilities=(0.4,0.2,0.2,0.3,0.3), sum=6.543112 and mean=1.308622.

Know the positional identity sin(i)sin(j)+cos(i)cos(j)=cos(i-j). With learned projections and content, the full attention score need not depend only on the offset. Computable encodings at longer positions do not guarantee accurate extrapolation.

Practice midterm / questions 1-4

Original practice, not a forecast of the actual paper / 40 marks / 60 minutes total

Use natural logs. Show formulas, substitutions, intermediate values and a one-sentence interpretation. Suggested timing: Q1 5 min; Q2 9 min; Q3 9 min; Q4 12 min; Q5 12 min; Q6 13 min. Do not open the solutions until time ends.

Q1. Foundations [4 marks]

(a) Distinguish p(y|x), p(x), and p(x|y)p(y). State how a generative classifier obtains a posterior. [2]
(b) P(spam)=0.3, P(win|spam)=0.7, P(win|ham)=0.1. Compute P(spam|win). [2]

Q2. AE/VAE [6 marks]

(a) Why is sampling an arbitrary latent point less reliable in a plain AE? How does a VAE address this? [2]
(b) A diagonal VAE encoder gives mu=(1,0), logvar=(ln 4, ln 0.25), and epsilon=(0.5,-2). Compute standard deviations and z. [2]
(c) Compute KL(q||N(0,I)). If the reconstruction negative log-likelihood is 0.2, report the negative ELBO with unit KL weight. [2]

Q3. GAN losses and gradient flow [6 marks]

(a) Write the saturating and non-saturating generator losses. Let D(G(z))=sigmoid(a); derive both derivatives with respect to a and evaluate them at d=0.01. [3]
(b) During a generator update, why must gradients pass through D even when D weights are frozen? What happens if you detach G(z)? [2]
(c) A generator makes convincing copies of a few faces for many different z. Name the failure and one lecture remedy. [1]

Q4. Metrics [8 marks]

(a) Three generated images have classifier probabilities (0.9,0.1), (0.5,0.5), and (0.1,0.9). Compute the marginal, each KL, and IS. [3]
(b) mu_r=(0,1), mu_g=(2,-1), S_r=diag(1,9), S_g=diag(4,1). Calculate the mean and covariance contributions and total FID. [3]
(c) One-dimensional real features are {0,2,10,12}; generated features are {0,1,2,3}. With k=1, excluding self and including boundaries, list the radii and calculate precision/recall. [2]

Before turning the page

For any short conceptual answer, give a definition followed by the mechanism or consequence. For numericals, avoid jumping straight to a calculator answer: the intermediate steps demonstrate that you chose the right distribution, direction, dimensions and reduction.

Practice midterm / questions 5-6

Continue the same 60-minute timer / These questions complete the 40-mark practice paper

Q5. Diffusion [8 marks]

(a) Explain why a variance-preserving step uses sqrt(1-beta_t) and sqrt(beta_t), assuming independent unit-variance inputs. [2]
(b) Let beta_1=0.2, beta_2=0.2, x_0=1, and aggregate epsilon=-0.5. Compute alpha_1, alpha_2, alpha_bar_2 and x_2 using the closed form. [2]
(c) A reverse step has x_t=0.8, alpha_t=0.8, alpha_bar_t=0.64, predicted noise=0.3, sigma_t=0.1 and z=-1. Compute mu_theta and x_(t-1). What changes for the final step? [2]
(d) State the noise-prediction training input and target. Why can training sample random timesteps directly? [2]

Q6. Transformers [8 marks]

(a) One query q=(1,0), keys k1=(1,0), k2=(0,1), values v1=(2,0), v2=(0,4), and d_k=2. Compute scaled scores, softmax weights, and the attention output. Then compute the output if key 2 is a masked future token. [3]
(b) For d_model=4 and pos=1, compute all four sinusoidal positional coordinates using base 10000. State whether to use degrees or radians. [2]
(c) Correct-token probabilities at three target positions are (0.5,0.25,0.8). Compute summed and mean cross-entropy. [2]
(d) State where Q, K and V come from in decoder cross-attention. [1]

Additional oral checks after marking

1. Why does identical p(y|x) across every generated image imply IS=1?
2. Can equal means and covariances prove the generator has learned the complete data distribution?
3. How do low recall and low precision imply different generation failures?
4. Why does the forward one-shot formula not eliminate the reverse generation loop?
5. Is the final reverse step stochastic in the basic DDPM algorithm?
6. Why must a causal mask be applied before softmax?
7. How does shifted teacher forcing avoid giving the decoder its current target as input?
8. Can non-saturating loss alone solve mode collapse?

If you have extra minutes, derive the diffusion reverse mean from the posterior coefficients on page 6, or redo the full robot/cross-attention examples directly from the slides. These give more coverage than another passive reread.

Solutions / questions 1-4

Self-mark against the allocated subpart marks / Diagnose method errors before arithmetic

Q1. Generative classification

(a) p(y|x) is the label posterior; p(x) is the observation distribution; p(x|y)p(y)=p(x,y). Normalize joint scores over classes to obtain p(y|x). (b) Spam score=0.7 x 0.3=0.21; ham score=0.1 x 0.7=0.07; posterior=0.21/(0.21+0.07)=0.75.

Q2. VAE

(a) Plain AE reconstruction does not constrain unused latent regions to decode plausibly. VAE learns a distribution and regularizes it toward a chosen prior, encouraging usable latent sampling. (b) Variances=(4,0.25), standard deviations=(2,0.5), so z=(1,0)+(2,0.5)*(0.5,-2)=(2,-1).

```
KL = 0.5[(1+4-1-ln 4)+(0+0.25-1-ln 0.25)]
= 0.5[2.61370564 + 0.63629436] = 1.625
Negative ELBO = 0.2 + 1.625 = 1.825
```

Q3. GAN

(a) L_sat=ln(1-sigmoid(a)); derivative=-sigmoid(a)=-d=-0.01. L_NS=-ln sigmoid(a); derivative=-(1-d)=d-1=-0.99. Their magnitudes differ by a factor of 99 at this d. If also checking values: L_sat=-0.01005034, L_NS=4.60517019.

(b) Chain rule needs dL/dG(z) through D, followed by dG(z)/d(theta_G). Freezing D prevents its weights updating, while preserving this input derivative. Detaching G(z) blocks generator gradients. (c) Mode collapse; minibatch discrimination or batch-diversity features are a lecture remedy.

Q4. Metrics

(a) Marginal=(0.5,0.5). KL1=0.9 ln(1.8)+0.1 ln(0.2)=0.36806421. KL2=0; KL3=0.36806421. Mean=0.24537614; IS=exp(0.24537614)=1.27810197. Confident predictions span both classes, but one uncertain image lowers the score.

(b) Squared mean distance=(-2)^2+(2)^2=8. Diagonal covariance term=(sqrt(1)-sqrt(4))^2+(sqrt(9)-sqrt(1))^2=1+4=5. FID=13. Equivalently, S_r S_g=diag(4,9) and its root=diag(2,3); trace contribution=5.

(c) Real radii are (2,2,2,2); generated radii are (1,1,1,1). All four generated queries lie in real balls, so precision=4/4=1. Real points 0 and 2 are in generated balls, but 10 and 12 are not: recall=2/4=0.5. Generated samples cover only the left real mode despite full precision.

Method checklist

Award yourself method credit only when the relevant steps are present: normalization in Bayes; converting log variance; the correct KL direction and sign; derivatives with respect to the named variable; marginal before KL; square-rooted variances in diagonal FID; own-set radii and correct denominators in precision/recall. This is a study rubric, not the instructor's marking scheme.

Solutions / questions 5-6

Finish by repairing the calculations you could not do closed-book

Q5. Diffusion

(a) Independent variance adds: Var(a x+b epsilon)=a^2+b^2. With noise variance beta, choose b=sqrt(beta) and a=sqrt(1-beta). The result has variance 1 under the stated unit-variance assumption. (b) alpha_1=alpha_2=0.8, alpha_bar_2=0.64; x_2=0.8 x 1+0.6 x (-0.5)=0.5.

```
(c) mu = [0.8 - (0.2/0.6)0.3]/sqrt(0.8)
= 0.7/0.89442719 = 0.78262379
x_(t-1) = 0.78262379 + 0.1(-1) = 0.68262379
```

At t=1 omit the added-noise term and output the mean. (d) Input=(x_t,t); target=aggregate epsilon used to construct x_t from x_0. The closed-form q(x_t|x_0) lets training draw a selected timestep directly instead of simulating all preceding steps; sampling timesteps estimates the training expectation across noise levels.

Q6. Transformers

```
(a) raw scores=(1,0); scaled=(0.70710678,0)
weights=(0.66976155,0.33023845)
output=0.66976155(2,0)+0.33023845(0,4)
=(1.33952310,1.32095380)
```

If key 2 is masked, its score becomes -infinity, weights become (1,0), and output becomes (2,0). Masking after softmax without renormalization would give the wrong result.

(b) i=0 uses denominator 1; i=1 uses 10000^(2/4)=100. PE=(sin 1,cos 1,sin 0.01,cos 0.01)=(0.84147098,0.54030231,0.00999983,0.99995000), in radians. (c) CE_sum=-ln(0.5)-ln(0.25)-ln(0.8)=2.30258509; CE_mean=0.76752836. (d) Q comes from decoder states, while K and V come from encoder outputs.

Twenty-minute repair plan

First 5 minutes: mark every missed subpart as formula selection, interpretation, matrix shape, sign/reduction, or arithmetic. Next 10 minutes: redo the two weakest numerical families from a blank page. Final 5 minutes: recite GAN gradients; Gaussian KL; IS/FID directions; forward/reverse diffusion; Q/K/V sources and causal masking.

Sources and limits

Read against the seven PDFs in courses/first sem/genai: Lecture1 Introduction.pdf (41 pages); Lecture 2 AE and VAE v2.pdf (71); Lecture GANs Part 1.pdf (70); Lecture GANs Part 2.pdf (97); Lecture Diffusion Models Part 1.pdf (34); Lecture Diffusion Models Part 2.pdf (46); Transformer.pdf (68). Total: 427 PDF pages, including repeated builds. Locators in this pack are PDF page positions, not printed slide numbers.

Quiz evidence: C:/Users/pc/Downloads/Quiz 1 MSDS solution.pdf, 3 PDF pages. Q1 and Q2 verified against their source pages; numerical answers in this pack were independently recomputed. Existing handwritten-note indexes were used to locate revision material. No actual midterm paper or instructor blueprint was supplied. Priority is a preparation judgment; the practice paper is newly authored and has no confirmed predictive status.